6 **(a)** A small coil of wire is situated in a non-uniform magnetic field, as shown in Fig. 6.1.

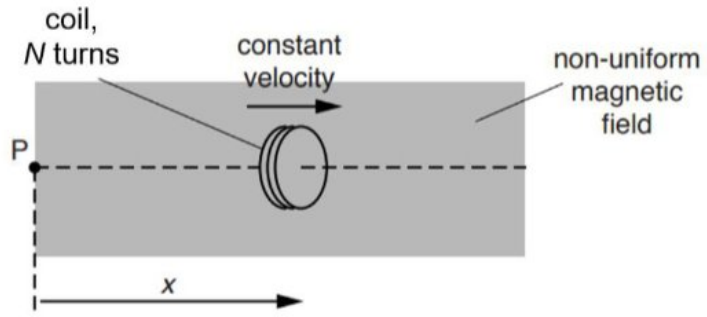


Fig. 6.1

The coil consists of N turns of wire and moves with a constant speed in a straight line.

The coil has displacement x from a fixed point P.

The variation with x of the e.m.f. E induced in the coil for values of x from $x = 0$ to $x = 6.0$ cm is shown in Fig. 6.2.

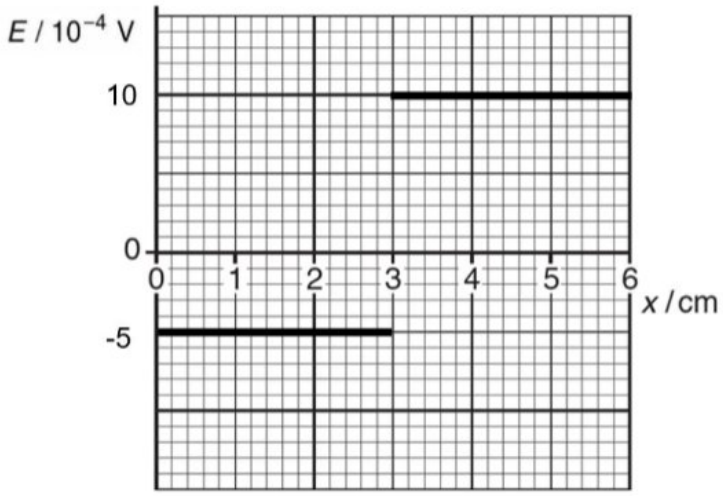


Fig. 6.2

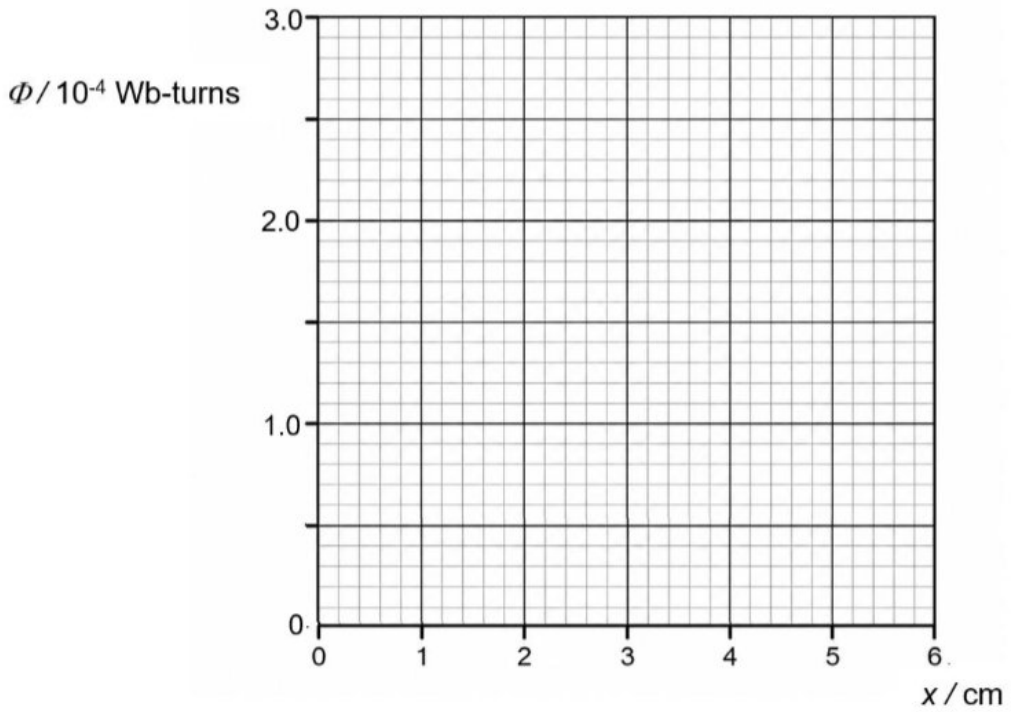
(i) The increase in the magnetic flux linkage of the coil is 8.0×10^{-5} Wb-turns when the coil moves from point P to the point where $x = 3.0$ cm. Using Fig. 6.2, determine the speed of the coil.

speed = ... m s^{-1} [2]

(ii) The magnetic flux at P is 3×10^{-6} Wb and the magnetic flux at $x = 3$ cm is 5×10^{-6} Wb. Determine N, the number of turns in the coil.

N = ... turns [1]

(iii) On Fig. 6.3, sketch the variation with x of the magnetic flux linkage Φ for values of x from $x = 0$ cm to $x = 6.0$ cm. [2]



(b) A metal ring is suspended from a fixed point P by means of a thread, as shown in Fig. 6.4.

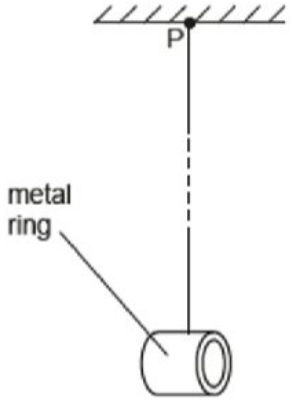


Fig. 6.4

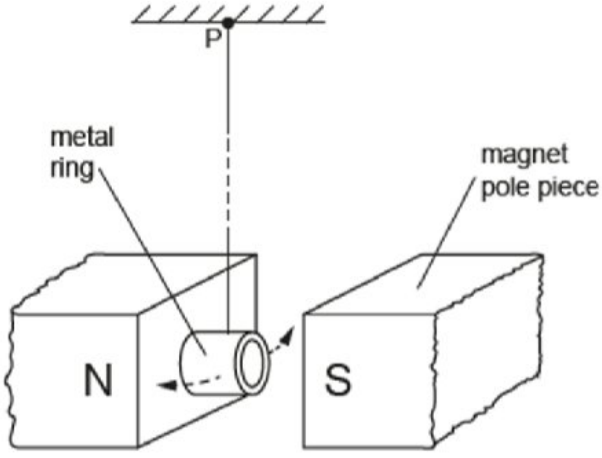


Fig. 6.5

The ring is displaced a distance *d* and then released. The ring completes many oscillations before coming to rest.

The poles of a magnet are now placed near to the ring so that the ring hangs midway between the poles of the magnet, as shown in Fig. 6.5.

The ring is again displaced a distance *d* and then released.

Explain why the ring completes fewer oscillations before coming to rest.

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..... [4]